

DP / NICU
22 1327

CRM NO 210643000174	Name B/o. ANJU	Father Name BALACHANDRAN	Lab ID 30500101091
Package NBS Trio	Gender MALE	DOB/Age 22/04/2023 (7 Days 1 Hrs)	Referred By Dr. SURESH
Hospital name SKS Hospital	City SALEM	Pre Term / Full Term Pre term	Baby Weight (Kg) 2.600
Specimen Source Venous Blood and Urine	Specimen Notes Acceptable	Collection Date & Time 30/04/2023 00:00	Received Date & Time 03/05/2023 14:20
Date of Report & time 04/05/2023 14:50	Blood Transfusion NO	Transfusion Date NO	Special Feeds / IVF / TPN/ Supplements Yes

INBORN ERRORS OF METABOLISM (IEM) SUMMARY REPORT

Sr. No.	Test Methodology	Result	Test Type	Page
1	FLUOROMETRIC ENZYME IMMUNOASSAY SCREENING REPORT	Negative	Screening	2
2	TANDEM MASS SPECTROMETRY SCREENING REPORT	Negative	Screening	3
3	GAS CHROMATOGRAPHY MASS SPECTROMETRY SCREENING REPORT	Negative	Screening	6

DISCLAIMER

All Investigations have their limitations which are imposed by the limits of sensitivity and specificity of individual assay procedures as well as the quality of specimen received by the laboratory.

It is presumed that the received specimen belongs to the patient named or Identified in the test request form.

Partial reproduction of this report is not permitted.

It is important to monitor baby's health and take a proactive approach should there be any unhealthy condition(s) within or beyond scope of this test.

Negative screening assay result does not rule out the possibility of an underlying metabolic / genetic disease.

The metabolites and health conditions mapping is not mutually unique and there can be other forms in which an abnormality may be apparent. Similarly, the health condition may not exclusively be an outcome of the disorders in consideration.

If the newborn is premature, retest is suggested when adjusted gestational age is 40 weeks.

HB-Variant hemoglobins may require further testing by the laboratory to diagnose.

HB-Presence of Hb Bart's with any newborn screen result indicates the presence of alpha thalassemia of unknown severity.

HB-If an increased amount of adult hemoglobin is noted, the infant is presumed to have been transfused (including Intrauterine transfusion).

HB-Carriers of beta thalassaemia and Hb Lepore cannot be excluded at this age.

Investigations performed is not intended to replace the medical advice and or treatment obtained from a qualified healthcare practitioner as it does not cover all the metabolic disorders

HB (Covered under Hemoglobinopathies package)

Dr. Mahesh Hampe

Dr. Mahesh Hampe, MD
Lab Director

CAP
ACCREDITED
COLLEGE OF AMERICAN PATHOLOGISTS

CRM NO : 210643000174

NAME : B/o. ANJU

REPORT - SUMMARY

Observations:

All the analytes are within the normal range.

Interpretation:

The tested sample is screened **NEGATIVE** for the list of disorders tabulated in page 2.

Comment:

It is recommended to repeat the test after 14 days in view of prematurity.

Recommendation:

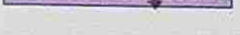
Please correlate the results with other clinical and diagnostic findings.

Initial Report ☒

Duplicate Report ☐

Revision Report ☐

Version No

Disorder	Parameter	Result*	Result Value	Reference	Remarks
Glucose 6 Phosphate Dehydrogenase deficiency	G6PD		100.6	NLT 3 U/dL	Normal
Classical Galactosemia	T GAL		3.1	NMT 25 mg/dL	Normal
Cystic Fibrosis	IRT		16.9	NMT 90 ng/mL	Normal
Congenital Hypothyroidism	TSH**		5.962	NMT 15 µU/mL	Normal
Congenital Adrenal Hyperplasia	17-OHP		14.43	NMT 55 ng/mL	Normal
Profound Biotinidase deficiency	BTD		170.9	NLT 40 U	Normal

* This Graph represents the value corresponding to results values. Each box represents the reference value. Results value is shown as dot.

** NMT - Not More Than, NLT - Not Less Than. ** TSH values are represented in Serum unit (Conversion factor: 1 µU/mL blood = 2.22 µU/mL serum).

G6PD - Glucose-6-Phosphate Dehydrogenase, T GAL - Total Galactose (galactose and galactose-1-phosphate), IRT - Human Immunoreactive Trypsin(ogen), TSH** - Thyroid Stimulating Hormone, 17-OHP - 17α-OH-progesterone, BTD - Profound Biotinidase

REFERENCES

- <https://www.aap.org/en-us/advocacy-and-policy/aap-health-initiatives/PEHDIC/Pages/Newborn-Screening.aspx>
- Sudhanshu S, Riaz I, Sharma R, Desai MP, Parikh R, Bhatia V. Newborn Screening Guidelines for Congenital Hypothyroidism In India: Recommendations of the Indian Society for Pediatric and Adolescent Endocrinology (ISPAE) – Part II: Imaging, Treatment and Follow-up. The Indian Journal of Pediatrics. 2018 Jun 1;85(6):448–53.

** End Of Report **

CRM NO : 210643000174

NAME : B/o. ANJU

REPORT- SUMMARY**Observations:**

All the analytes are within the normal range.

Interpretation:The tested sample is screened **NEGATIVE** for the list of disorders in Annexure 1.**Comment:**

It is recommended to repeat the test after 14 days in view of prematurity.

Recommendation:

Please correlate the results with other clinical and diagnostic findings.

CRM NO : 2105643000174

NAME : B/o. ANJU

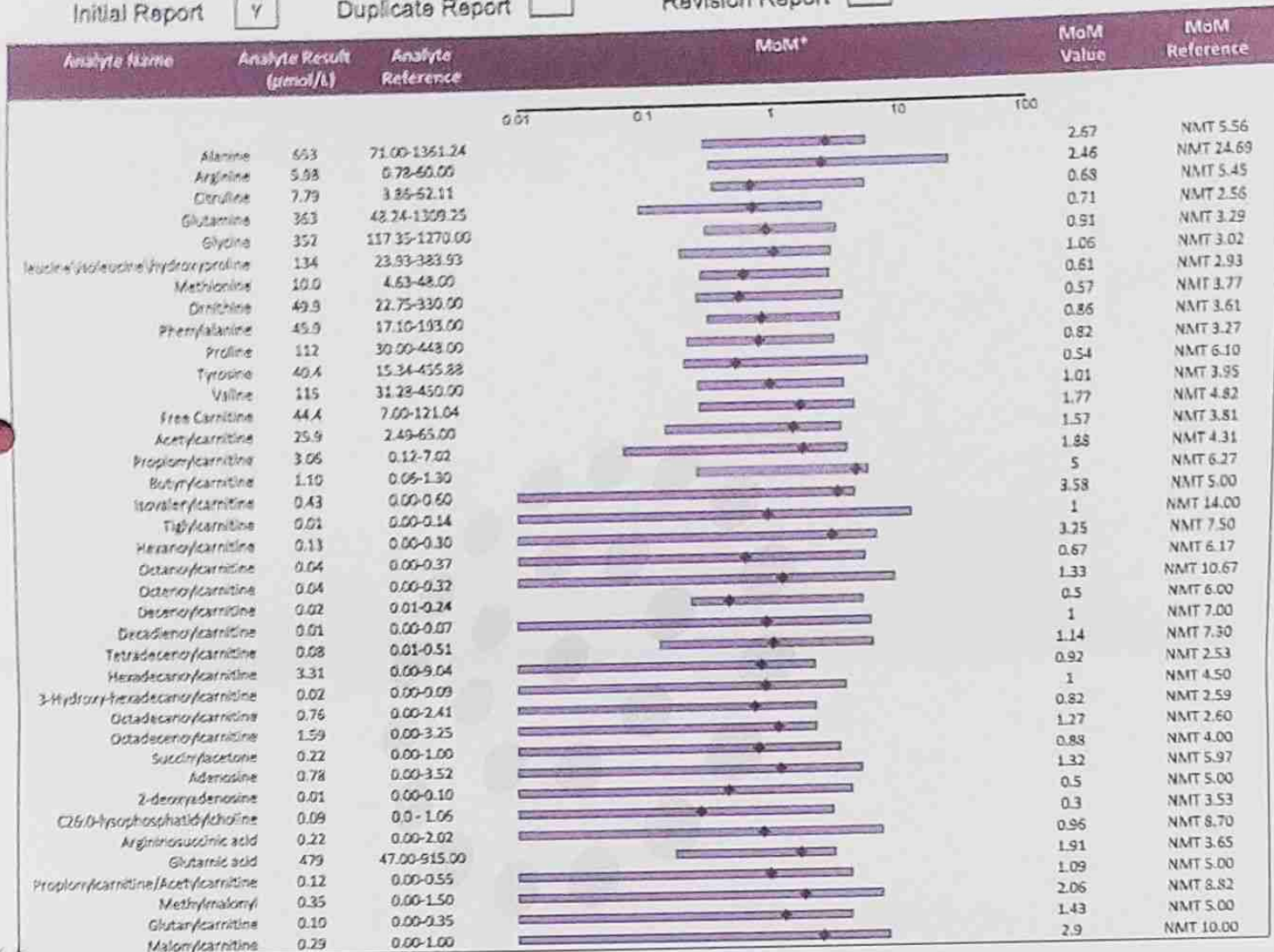
Initial Report ☒

Duplicate Report ☐

Revision Report ☐

Version No

1



* This Graph represents the value corresponding to Multiple of Median (MoM). Each box represents the interval between 1st and 99th percentile. The MoM Shown as dot.

CRM NO : 210643000174

NAME : B/o. ANJU

ANNEXURE - 1 LIST OF DISORDERS

Amino Acid Disorders	Organic Acid Disorders
Citrullinemia Type I	Long chain L-3 hydroxyacyl-CoA dehydrogenase deficiency
Citrullinemia Type II	Malonic acidemia
Hyperornithinemia with Gyral Atrophy	Methylmalonic acidemia (Cobalmin Disorders)
5-Oxoprolinuria	Methylmalonic acidemia (Mutase)
Transient Neonatal Tyrosinemia	3-Methylcrotonyl-CoA carboxylase deficiency
Argininosuccinic Acidemia	Multiple Carboxylase Deficiency (also known as Holocarboxylase synthase deficiency)
Argininemia	3-Methylglutaconic aciduria
Defects of bipterin cofactor biosynthesis	2-Methyl-Butyryl-Glycinuria
Disorders of bipterin cofactor regeneration	2-Methyl-3-Hydroxybutyric Aciduria
Benign Hyperphenylalaninemia	Propionic acidemia
Homocystinuria	Other Conditions
Hypermethioninemia	Adenosine deaminase (ADA)-severe combined
Maple syrup (urine) disease	Carbamoyl phosphate synthetase I
Non-ketotic hyperglycinemia	Hyperornithinemia-Hyperammonemia-Homocitrullinuria
Phenylketonuria	Hyperalimentation (TPN)
Tyrosinemia type I	Liver Complications
Tyrosinemia type II	Medium Chain Triglyceride (MCT) Oil Administration
Tyrosinemia type III	N-acetylglutamate synthase deficiency
Fatty Acid Oxidation Disorders	Ornithine transcarbamylase deficiency
Carnitine Palmitoyl Transferase Type I Deficiency	Treatment with antibiotics/Pyruvic Acid
Carnitine Uptake Defect	Treatment with Benzoate or Valproic Acid
Carnitine Acyl-Carnitine Translocase deficiency	X-linked adrenoleukodystrophy
Carnitine Palmitoyl-Transferase type II deficiency	
2,4 Dienoyl-CoA reductase deficiency	
Glutaric acidemia type II	
Medium-Chain Acyl-CoA Dehydrogenase Deficiency	
Medium-chain ketoacyl-CoA thiolase deficiency	
Medium/short-chain L-3-hydroxyacyl-CoA dehydrogenase deficiency	
Short-Chain Acyl-CoA Dehydrogenase Deficiency	
Trifunctional protein deficiency	
Very long-chain acyl-CoA dehydrogenase deficiency	
Organic Acid Disorders	
Isobutyryl-CoA-Dehydrogenase Deficiency	
Glutaric Acidemia type I	
3-Hydroxy-3-Methyl-Glutaric aciduria	
Iso-Valeric Acidemia	
Beta-KetoThiolase deficiency	

REFERENCES

1. Ombrone D, Giocallere E, Forni G, Malvagla S, la Marca G. Expanded newborn screening by mass spectrometry: New tests, future perspectives. Mass Spectrom Rev. 2016 Feb;35(1):71-84.
2. Gaugler S, Rykl J, Wegner I, Von Däniken T, Fingerhut R, Schlotterbeck G. Extended and Fully Automated Newborn Screening Method for Mass Spectrometry Detection. International Journal of Neonatal Screening. 2018 Mar;4(1):2.

** End Of Report **

CRM NO : 210643000174

NAME : B/o. ANJU

REPORT- SUMMARY**Observations:**

All the analytes are within the normal range.

Interpretation:The tested sample is screened **NEGATIVE** for the list of disorders in Annexure 1.**Recommendation:**

Please correlate the results with other clinical and diagnostic findings.

CRM NO : 210643000174

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Initial Report

☒ Y

Duplicate Report

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Revision Report

☐

Version No

1

Analyte Name	Analyte Result (nmol/mol of creatinine)	Analyte Reference	Log of MoM*	MoM Value	MoM Reference
2-Deoxytetronic acid-3	194.15	NMT 806.88		0.29	NMT 0.91
2-Hydroxyadipic acid-3	9.73	NMT 479.95		0.1	NMT 1.79
2-Hydroxyglutaric acid-3	128.01	NMT 433.03		0.43	NMT 0.96
2-Hydroxyisovaleric acid-2	2.33	NMT 208.50		0.1	NMT 2.05
2-Ketoglutaric acid-3	34.75	NMT 4941.80		-0.72	NMT 1.44
2-Methyl-3-hydroxybutyric acid-2	6.32	NMT 569.81		-0.13	NMT 1.83
3-Hydroxy-3-methylglutaric acid-3	17.97	NMT 152.82		0.12	NMT 1.05
3-Hydroxyadipic acid-3	39.48	NMT 597.47		-0.01	NMT 1.17
3-Hydroxybutyric acid-2	147.13	NMT 25913.94		0.25	NMT 2.50
3-Hydroxyisovaleric acid-2	38.71	NMT 330.38		0.56	NMT 1.49
3-Hydroxypropionic acid-2	21.27	NMT 473.32		0.1	NMT 1.45
3-Hydroxysebacic acid-3	6.62	NMT 298.69		-0.24	NMT 1.41
3-Methylglutaconic acid(E)-2	10.42	NMT 271.90		0.09	NMT 1.51
3-Methylglutaconic acid(Z)-2	5.81	NMT 101.54		-0.23	NMT 1.01
3-Methylglutaric acid-2	2.47	NMT 185.66		-0.25	NMT 1.62
4-Hydroxyphenylacetic acid-2	22.57	NMT 773.55		-0.15	NMT 1.39
4-Hydroxyphenyllactic acid-3	145.61	NMT 2583.12		1.1	NMT 2.35
4-Hydroxyproline-3	90.21	NMT 750.91		0.79	NMT 1.71
5-Hydroxy Caproic Acid-2	11.39	NMT 138.86		0.19	NMT 1.28
5-Oxoproline-2	76.97	NMT 704.71		0.46	NMT 1.42
Adipic acid-2	4.84	NMT 296.01		-0.97	NMT 1.42
Alanine-2	121.59	NMT 1083.58		0.28	NMT 1.23
Cysteine-3	8.57	NMT 373.99		-0.57	NMT 1.06
Ethylhydracrylic acid-2	6.33	NMT 138.32		-0.12	NMT 1.22
Ethylmalonic acid-2	27.05	NMT 228.15		0.35	NMT 1.27
Fumaric acid-2	17.61	NMT 276.25		0.31	NMT 1.51
Galactitol-6	147.96	NMT 788.22		0.58	NMT 1.31
Galactose-5	1144.89	NMT 4053.36		0.81	NMT 1.36
Glutaric acid-2	7.98	NMT 116.88		0.37	NMT 1.54
Glyceric acid-3	35.67	NMT 424.53		0.28	NMT 1.36
Glycerol 3-phosphate-4	26.14	NMT 203.13		0.24	NMT 1.13
Glycerol-3	234.72	NMT 19571.15		0.12	NMT 2.04
Glycine-2	249.38	NMT 3251.01		0.16	NMT 1.28
Glycine-3	151.16	NMT 1722.91		0.36	NMT 1.42
Glycolic acid-2	129.24	NMT 1238.59		0.24	NMT 1.22
Homovanillic acid-2	17	NMT 316.91		0.29	NMT 1.57
Lactic acid-2	366.45	NMT 21683.55		0.19	NMT 1.96
Leucine-2	3.37	NMT 138.26		-0.08	NMT 1.54
Malic acid-3	31.83	NMT 413.27		0.5	NMT 1.61
Methylcitric acid-4	44.85	NMT 2014.08		-0.07	NMT 1.59
Methylcitric acid-4	2.4	NMT 24.34		0.39	NMT 1.40
Methylmalonic acid-2	3.44	NMT 1216.56		-0.44	NMT 2.10
N-Acetylaspartic acid-2	42.59	NMT 206.59		0.12	NMT 0.81
N-Acetyls erine-2	9.25	NMT 66.16		0.39	NMT 1.25
Orotic acid-3	3.83	NMT 43.80		0.38	NMT 1.44
Oxalic acid-2	27.01	NMT 2129.71		-0.18	NMT 1.72
Phenylalanine-2	3.43	NMT 211.64		-0.16	NMT 1.63
Pyruvic acid-2	183.46	NMT 1216.32		0.39	NMT 1.21
Sebacic acid-2	24.18	NMT 3508.91		0.04	NMT 2.20
Suberic acid-2	5.86	NMT 470.32		-0.39	NMT 1.52
Succinic acid-2	371.38	NMT 2057		0.76	NMT 13.35
Threonine-3	247.29	NMT 1249.40		0.99	NMT 1.69
Uric acid-4	1575.97	NMT 7823.44		0.38	NMT 1.07
Valine-2	6.6	NMT 302.83		-0.07	NMT 1.59
Vanillylmandelic acid-3	7.85	NMT 47.81		0.26	NMT 1.05
Methylsuccinic acid-2	2.7	NMT 52.51		-0.05	NMT 1.24

* This Graph represents the metabolites which are found above the detection limit. Each box represents the reference interval between 1%ile and 99%ile Multiple of Median(MoM) in Logarithmic scale. The Log of MoM of the analyzed sample is shown as dot.

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ANNEXURE - GCMS LIST OF DISORDERS

Amino Acidopathies and Organic Acidemias	Amino Acidopathies and Organic Acidemias	TCA Cycle/Mitochondrial Abnormality
2-Hydroxyglutaric aciduria	Hyperleucine-Isoleucinemia	Pyruvate dehydrogenase phosphatase deficiency
2-ketoadipic aciduria	Hyperglycinuria (Ketotic)	Disorders of Fatty Acid Metabolism
3-hydroxy-3-methylglutaryl-CoA-lyase deficiency	Hyperglycinuria (non-ketotic)	Dicarboxylic aminoaciduria
3-Hydroxyisobutyryl CoA Deacylase Deficiency	Hypermethioninemia	Ethyl Malonic Aciduria
3-methylcrotonyl CoA carboxylase deficiency	Hypersarcosinemia	Glycerol Kinase Deficiency
3-methylglutaconic aciduria Type I	Hyperlysinemia Type 1	Medium chain acyl CoA dehydrogenase deficiency
5-oxoprolinuria (also known as pyroglutamic aciduria)	Isobutyryl-CoA dehydrogenase deficiency (IBD)	Mitochondrial trifunctional protein deficiency
Alkaptonuria	Iminoglycinuria	Very Long-chain acyl-CoA dehydrogenase deficiency
Aminoacylase I Deficiency	Imidazole Amino Aciduria	Short chain acyl CoA dehydrogenase dehydrogenase deficiency
Argininemia	Isovaleric acidemia	Zellweger like syndrome
Arglinosuccinic aciduria	Lysinuric protein intolerance	Peroxisomal Disorders
Benign hyperphenylalaninemia	Malonic acidemia	Infantile rebus disease
Beta-Aminoisobutyric Aciduria	Maple syrup urine disease (MSUD)	Neonatal Adrenoleukodystrophy
Profound Biotinidase deficiency	Methylmalonic aciduria; cblA and cblB forms	Primary hyperoxaluria Type 1
Beta- ketothiolase deficiency (BKT)	Methylmalonic acidemia and Homocystinuria -Cbl C, D	Primary hyperoxaluria Type 2
Canavan disease	Methylmalonic semialdehyde dehydrogenase deficiency	Zellweger syndrome
Carbamoyl Phosphate Synthetase-1 Deficiency / N-Acetylglutamate Synthetase Deficiency	Methylmalonyl- CoA mutase deficiency (MUT)	Disorders of Purine and Pyrimidine Metabolism
Carnosinuria	Mevalonic aciduria	Adenine phosphoribosyl transferase deficiency
Citrullinemia Type 1	Multiple carboxylase deficiency	Dihydropyrimidinase Deficiency
Citrullinemia type II	N-Acetylglutamate Synthetase Deficiency (merged with CPS1)	Hyperuric acidemia
Cystinuria	NICCD	Lesch-Nyhan syndrome
Cystathioninuria	Ornithine Trans Carbamylase Deficiency	Molybdenum cofactor deficiency
Defects of bipterin cofactor regeneration (BIOPT REG)	Phenylketonuria (PKU)	Orotic aciduria
Dihydrolipoyl Dehydrogenase (E3) Deficiency (also known as MSUD Type 3)	Propionic acidemia	Partial deficiency of hypoxanthine-guanine phosphoribosyltransferase
Familial Renal Iminoglycinuria	Saccharopinuria (also known as Hyperlysinemia Type II)	Thymine Uraciluria
Formiminoglutamic aciduria	Succinic semialdehyde dehydrogenase deficiency	Xanthinuria (Type I & II)
Glutaric aciduria type II	Tyrosinemia caused by liver dysfunction	Disorders of Sugars
Glutaric aciduria type I	Transient neonatal tyrosinemia	D-glyceric aciduria
Glutathionuria	Tryptophanuria with dwarfism	Endogenous sucrosuria
Hartnup Disease	Tyrosinemia Type II	Fructose-1 and 6-diphosphatase deficiency
Hawkinsinuria	Tyrosinemia Type III (also covers transient neonatal tyrosinemia)	Fructosuria
Hyperornithinemia-hyperammoninemia-hyperhomocitrullinemia (HHH) syndrome	Tyrosinemia Type I	Galactokinase deficiency (GALK)
Histidinuria	Valinemia	Galactose epimerase deficiency (GALE)
Histidinemia	Xanthurenic aciduria	Lactic Acidosis
Homocystinuria	TCA Cycle/Mitochondrial Abnormality	Transient Galactosemia
Hyperprolinemia type I	Fumarate hydratase deficiency	Non-IEM Disorder
Hyperprolinemia type II	Leigh syndrome	Neuroblastoma
Hydroxylysinauria	Pyruvate carboxylase deficiency	
Hyperhydroxyprolinemia	Pyruvate decarboxylase deficiency	
	Pyruvate dehydrogenase (E1) deficiency	

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REFERENCES

1. Ombrone D, Giocellere E, Forni G, Malvagla S, Ia Marca G. Expanded newborn screening by mass spectrometry: New tests, future perspectives; Mass Spectrom Rev. 2016 Feb;35(1):71-84.
2. Gaugler S, Rykl J, Wegner I, Von Däniken T, Fingerhut R, Schlötterbeck G. Extended and Fully Automated Newborn Screening Method for Mass Spectrometry Detection. International Journal of Neonatal Screening. 2018 Mar;4(1):2.

**** End Of Report ****